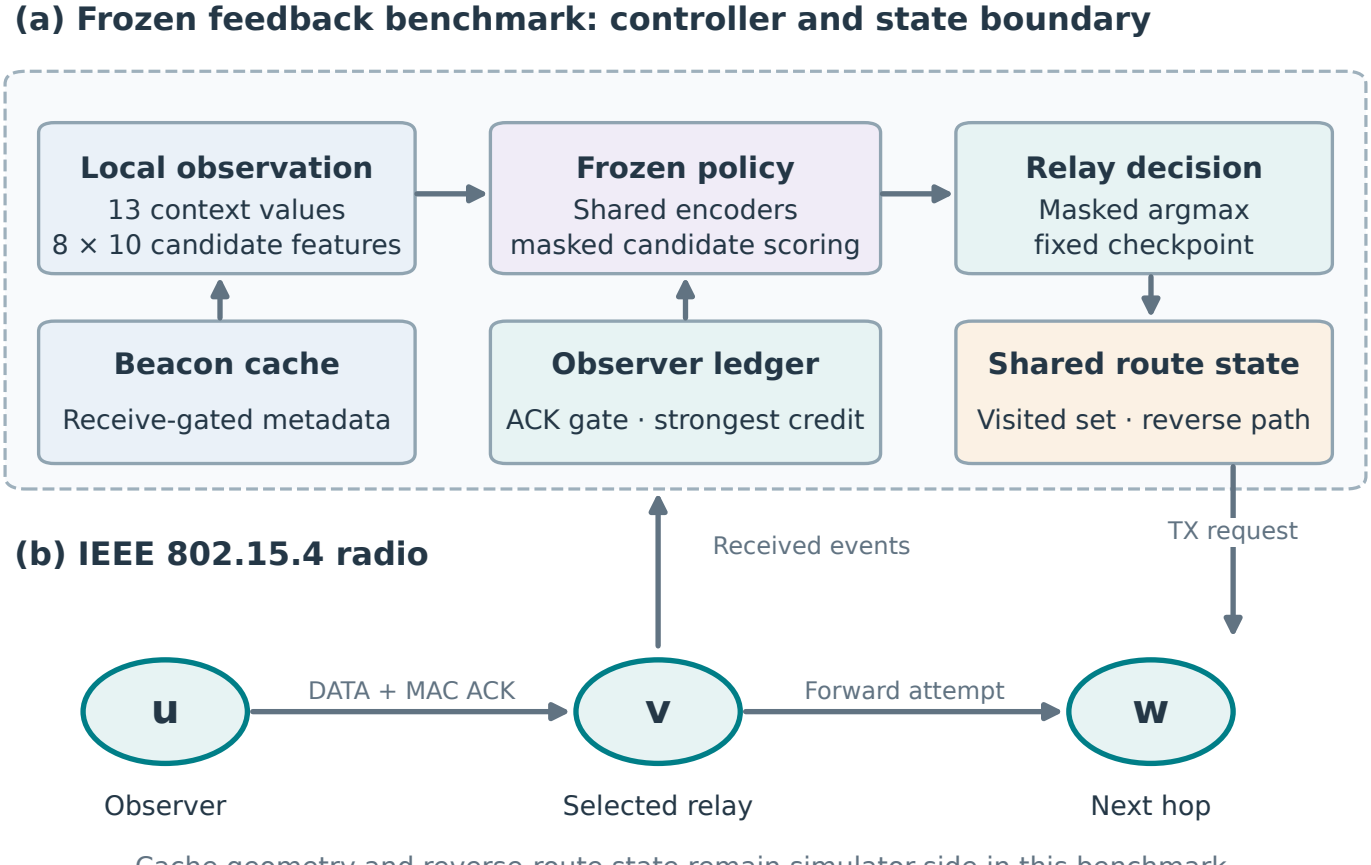
(a) Frozen feedback benchmark: controller and state boundary
Local observation
13 context values
8 × 10 candidate features
Frozen policy
Shared encoders
masked candidate scoring
Relay decision
Masked argmax
fixed checkpoint
Beacon cache
Receive-gated metadata
Observer ledger
ACK gate · strongest credit
Shared route state
Visited set · reverse path
Received events
TX request
(b) IEEE 802.15.4 radio
u
DATA + MAC ACK
v
Forward attempt
w
Observer
Selected relay
Next hop
Cache geometry and reverse-route state remain simulator-side in this benchmark.